TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 18

5-MINUTE REVISION

# Analysis of Variance

Partitioning total variation, degrees of freedom, mean squares, the F test, assumptions and ANOVA models.

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STAT 3202 · Chapter revision

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THE IDEA TO KEEP

**ANOVA compares means through a variance ratio.**

**REMEMBER**

A significant treatment F test indicates that not all treatment means are equal. It does not identify the differing pairs.

**APPLY IT**

For three treatments and 12 observations in a one-way CRD, treatment df = 2, error df = 9 and total df = 11.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

## 1 The idea

**ANOVA** (Analysis of Variance) compares population means by **partitioning the total variation** into parts due to known sources and to error.

Null hypothesis (one-way)

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_v$$

A significant $F$ means **at least one pair** of means differs. It is an **omnibus** test and doesn't say which pairs.

## 2 Assumptions

- ✓ Observations from **normal** populations
- ✓ Common variance: **homogeneity of variance**
- ✓ Samples drawn **independently**
- ✓ Treatment effects are **additive**

So errors are iid $N(0, \sigma^2)$. Check with residual plots: residuals vs fitted values, and a normal Q-Q plot.

## 3 One-way ANOVA (CRD)

Correction factor

$$CF = \frac{G^2}{n}$$

Total SS

$$SS_T = \sum y_{ij}^2 - CF$$

Treatment SS

$$SS_{Tr} = \sum \frac{T_i^2}{n_i} - CF$$

Error SS

$$SS_E = SS_T - SS_{Tr}$$

Mean squares

$$MS = \frac{SS}{df}$$

F ratio

$$F = \frac{MS_{Tr}}{MSE}$$

$v$ treatments, $n$ observations, $T_i$ treatment totals, $G$ grand total.

## 4 Degrees of freedom

| Source | One-way | Two-way |
|---|---|---|
| Treatments | $v-1$ | $v-1$ |
| Blocks | – | $r-1$ |
| Error | $n-v$ | $(v-1)(r-1)$ |
| **Total** | $n-1$ | $vr-1$ |

## 5 Two-way ANOVA (RBD)

Block SS ($B_j$ block totals)

$$SS_B = \sum \frac{B_j^2}{v} - CF$$

Error SS

$$SS_E = SS_T - SS_{Tr} - SS_B$$

Separates treatment and block variation. With one observation per cell, interaction cannot be separated from error.

## 6 ANOVA models

Linear model

$$Y_{ij} = \mu + \tau_i + e_{ij}, \quad e_{ij} \sim N(0, \sigma^2)$$

| | |
|---|---|
| **Fixed** | $\tau_i$ unknown constants (specific varieties of interest) |
| **Random** | $\tau_i \sim N(0, \sigma_\tau^2)$; variance components (randomly chosen fields) |
| **Mixed** | Both fixed and random effects |

## 7 The ANOVA table

| Source | df | SS | MS | F |
|---|---|---|---|---|
| Treatments | $v-1$ | $SS_{Tr}$ | $SS_{Tr}/(v-1)$ | $MS_{Tr}/MSE$ |
| Error | $n-v$ | $SS_E$ | $MSE$ | |
| **Total** | $n-1$ | $SS_T$ | | |

Reject $H_0$ if calculated $F$ > table $F$ at ($v-1$, error df). Then use multiple comparisons to find which means differ.

## ! Common mistakes

- ✗ Concluding **all** means differ from a significant $F$.
- ✗ Forgetting to subtract the correction factor.
- ✗ Deleting outliers just to change significance.

## ? Fill in the blanks

1. A CRD with 5 treatments and 4 replications has ____ error df. [ 15 ]
2. An RBD with 5 treatments in 4 blocks has ____ error df. [ 12 ]
3. MS(treatment) = 40 and MSE = 8 give F = ____. [ 5 ]
4. A grand total of 120 from 12 observations gives a correction factor of ____. [ 1200 ]
5. A treatment SS of 60 with 4 treatments gives a treatment mean square of ____. [ 20 ]
6. A significant F shows that at least ____ pair of means differs. [ one ]